1 Week 5: General camera models and camera calibration

So far our camera calibration matrix has made a number of simple assumptions that may not be very realistic for all cameras. Below we briefly step through some modifications to handle this.

1.1 Non-square pixels

Some cameras have non-square pixels, which means the focal length is different in the u and v directions. We call the ratio of the different focal lengths f_u/f_v the *aspect ratio* of an image.

$$\begin{bmatrix} \lambda u \\ \lambda v \\ \lambda \end{bmatrix} = \begin{bmatrix} f_u & 0 & u_0 \\ 0 & f_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \tag{1}$$

1.2 Skew distortion

Sometimes, the pixels are skewed, and we need to factor this in using a skew parameter s.

$$\begin{bmatrix} \lambda u \\ \lambda v \\ \lambda \end{bmatrix} = \begin{bmatrix} f_u & s & u_0 \\ 0 & f_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \tag{2}$$

1.3 Radial distortion

Cameras often exhibit distortion, a bending effect that warps image. Radial distortion is particularly common, and is easily spotted when straight lines appear curved, as shown below.

Figure 1: Curvature of straight lines is a clear indication of radial distortion

We can model radial distortion using a warping equation:

$$\mathbf{r}' = \mathbf{r}(1 + \alpha \mathbf{r}^2) \tag{3}$$

where $\mathbf{r} = [u, v]^{\mathrm{T}}$. Intuitively this means that if α is positive, points further away from the image centre are pushed away even more, while if α is negative, points further are pulled in towards the origin more. The latter is shown below, and termed barrel distortion.

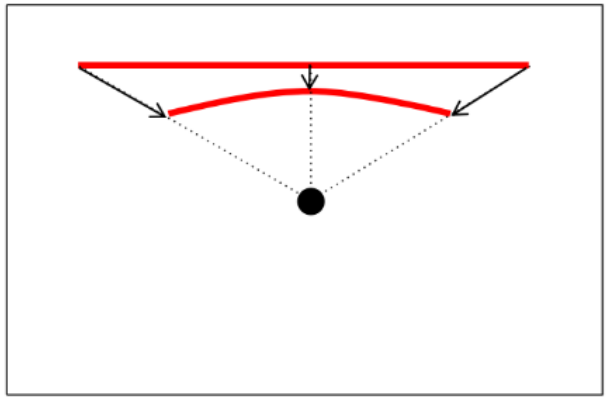

Figure 2: Barrel distortion is modelled using $\alpha < 0$

Sometimes this model isn't enough, and we use higher order models

$$\mathbf{r}' = \mathbf{r}(1 + \alpha\mathbf{r}^2 + \beta\mathbf{r}^4 + \ldots) \tag{4}$$

We can undistort an image, and correct for distortion by knowing that an image should contain straight lines. A typical camera calibration pipeline is as follows, use a known calibration object with easily detected points, and solve for the distortion parameters, alongside camera calibration parameters. These are then fixed for a given camera, so that images can be unwarped and treated as obeying the pinhole camera model for downstream applications.

Figure 3: Undistorting an image gives it the properties of a pinhole camera.

1.4 Decomposing a projection matrix

We now have a complete camera model:

$$\begin{bmatrix} \lambda u \\ \lambda v \\ \lambda \end{bmatrix} = \mathbf{P} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix} \tag{5}$$

$$\begin{bmatrix} \lambda u \\ \lambda v \\ \lambda \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix} \tag{6}$$

$$\begin{bmatrix}\lambda u\\ \lambda v\\ \lambda\end{bmatrix} = \begin{bmatrix} f_u & s & u_0\\ 0 & f_v & v_0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x\\ r_{21} & r_{22} & r_{23} & t_y\\ r_{31} & r_{32} & r_{33} & t_z\end{bmatrix}\begin{bmatrix} x_w\\ y_w\\ z_w\\ 1\end{bmatrix} \tag{7}$$

$$\begin{bmatrix}\lambda u\\ \lambda v\\ \lambda\end{bmatrix} = \mathbf{K}\begin{bmatrix}\mathbf{R} & \mathbf{t}\end{bmatrix}\begin{bmatrix} x_w\\ y_w\\ z_w\\ 1\end{bmatrix} \tag{8}$$

There are 11 degrees of freedom in this model (3 for rotation, 3 for translation, 5 for the camera). So the Projection matrix has 11 degrees of freedom, but 12 numbers (we don't care about the scale).

Interestingly, any $\mathbf{P}$ can be decomposed into a matrix $\mathbf{K}\begin{bmatrix}\mathbf{R} & \mathbf{t}\end{bmatrix}$ using $\mathbf{RQ}$ decomposition. This is useful if we have solved for the projection matrix and want to find intrinsic and extrinsic calibration matrices.

You may recall that $\mathbf{A} = \mathbf{QR}$ decomposition decomposes the matrix $\mathbf{A}$ into an upper triangular matrix $\mathbf{R}$ and an orthogonal matrix $\mathbf{Q}$ satisfying $\mathbf{Q}^{\mathrm{T}}\mathbf{Q} = \mathbf{I}$.

Looking at the projection matrix above we can see that the camera matrix $\mathbf{K}$ is an upper triangular matrix, and rotation matrix $\mathbf{R}$ is an orthogonal matrix (all rotations are), so to find the camera and transformation matrices, we want to do an $\mathbf{RQ}$ decomposition.

Let's look closely at the relationships below:

$$\begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14}\\ p_{21} & p_{22} & p_{23} & p_{24}\\ p_{31} & p_{32} & p_{33} & p_{34}\end{bmatrix} = \begin{bmatrix} f_u & s & u_0\\ 0 & f_v & v_0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x\\ r_{21} & r_{22} & r_{23} & t_y\\ r_{31} & r_{32} & r_{33} & t_z\end{bmatrix} \tag{9}$$

Multiplying out the bottom row of the equation tells us that $p_{31} = r_{31}$, $p_{32} = r_{32}$, $p_{33} = r_{33}$ and $p_{34} = t_z$. We also know that rows of $\mathbf{R}$ have magnitude 1, so

$$p_{31}^2 + p_{32}^2 + p_{33}^2 = 1. \tag{10}$$

Looking at the middle row, we obtain a relationship:

$$\begin{bmatrix} p_{21} & p_{22} & p_{23}\end{bmatrix} = f_v\begin{bmatrix} r_{21} & r_{22} & r_{23}\end{bmatrix} + v_0\begin{bmatrix} r_{31} & r_{32} & r_{33}\end{bmatrix} \tag{11}$$

We already know the second part, and we also know that the two rows of $\mathbf{R}$ are orthogonal with unit length. So we can solve for

$$v_0 = \begin{bmatrix} p_{21} & p_{22} & p_{23}\end{bmatrix}\cdot\begin{bmatrix} r_{31} & r_{32} & r_{33}\end{bmatrix} \tag{12}$$

by taking the projection of the projection matrix row onto the rotation row using the dot product.

We can then solve for the vector:

$$f_v\begin{bmatrix} r_{21} & r_{22} & r_{23}\end{bmatrix} = \begin{bmatrix} p_{21} & p_{22} & p_{23}\end{bmatrix} - v_0\begin{bmatrix} r_{31} & r_{32} & r_{33}\end{bmatrix} \tag{13}$$

Taking the magnitude of this gives us f_v, and we get $\begin{bmatrix} r_{21} & r_{22} & r_{23}\end{bmatrix}$ by normalising.

We can repeat this process for the final row, and successfully break our projection matrix into intrinsic and extrinsic components.

1.5 Vanishing points and lines

Projective geometry has interesting effects on various geometric objects. It's particularly interesting to explore what happens to points and lines under projection, as this introduces constraints that we can exploit for 3D reconstruction.

For example, when a square is viewed under a projection, parallel lines are no longer parallel, and meet at *vanishing points*. The set of vanishing points for parallel lines on a plane is called the vanishing line. Planes have a set of special *points at infinity* that project to the vanishing line.

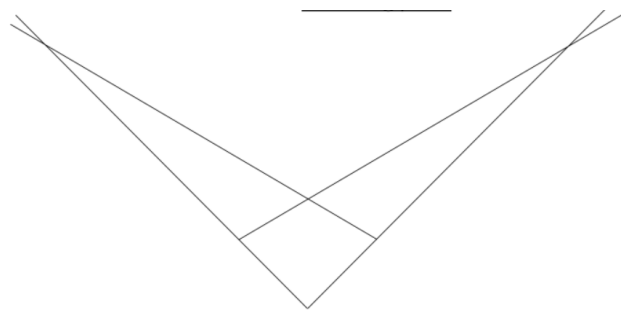

Figure 4: Images of parallel lines result in lines that meet at vanishing points.

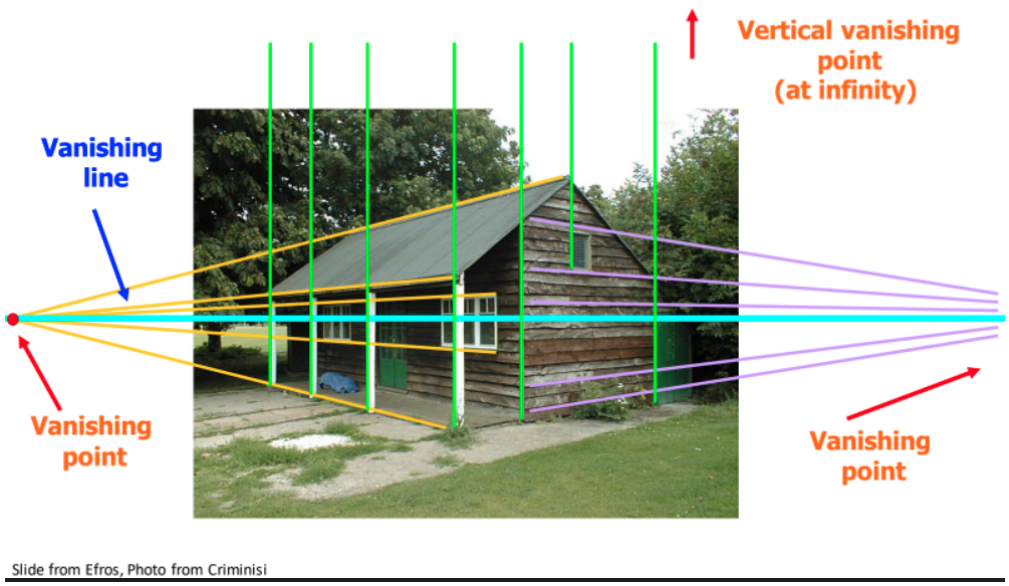

Figure 5: Vanishing points and lines (Efros, Photo from Criminisi).

Recall that we can represent a point x, y on a plane using a homogenous coordinate $[x, y, 1]^{\mathrm{T}}$. Points on the line at infinity take on the form $[x, y, 0]^{\mathrm{T}}$, we can't normalise to make them 1, but we know they lie at infinity in the direction x, y.

More generally, we can represent a line using the equation

$$ax + by + c = 0 \tag{14}$$

or as a dot product

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0 \tag{15}$$

Here, we are using the 3 dimensional vector $[a, b, c]^{\mathrm{T}}$. to represent the line of points for which the dot product is zero. The scale of this vector doesn't matter, because it's the same line, and the dot product is still zero regardless. Using this representation, we can think about the points at infinity:

$$\begin{bmatrix} x \\ y \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 0 \tag{16}$$

and horizontal or vertical lines

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \cdot \begin{bmatrix} a \\ 0 \\ c \end{bmatrix} = 0 \tag{17}$$

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ b \\ c \end{bmatrix} = 0 \tag{18}$$

We know that points on a plane transform when we apply a homography (note the equivalence relationship indicating up to scale)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \equiv \mathbf{H} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \tag{19}$$

so how do lines transform? Well, since a point on a line obeys the relationship

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0 \tag{20}$$

we expect

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0 \tag{21}$$

to be true too. Re-writing the dot product as a vector transpose

$$\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = 0 \tag{22}$$

and inserting $\mathbf{I} = \mathbf{H}^{-1}\mathbf{H}$

$$\begin{bmatrix} a & b & c \end{bmatrix} \mathbf{H}^{-1}\mathbf{H} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = 0 \tag{23}$$

we obtain

$$\begin{bmatrix} a & b & c \end{bmatrix} \mathbf{H}^{-1} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = 0 \tag{24}$$

This means that line vectors $l = [a, b, c]^{\mathrm{T}}$ are transformed as

$$\begin{bmatrix} a' \\ b' \\ c' \end{bmatrix} = \mathbf{H}^{-\mathrm{T}} \begin{bmatrix} a \\ b \\ c \end{bmatrix} \tag{25}$$

The line vector l has an interesting interpretation, as shown below:

We can use the line vector to quickly test if a point lies on the line - the dot product between the line and a point on it will be zero. Ok, how do we find the line vector? Easy, if we have two points on the line, we just take their cross product.

All of this can be expanded into three dimensions too, and we can derive similar relationships for circles and ellipses (or conics). This is particularly useful for model-based computer vision.

If we know what shape we are looking for, we can exploit these constraints as tests in ransac-like algorithms. For example, we can find lines by sampling point pairs and checking consensus along the line, as done in our pool table example in class.

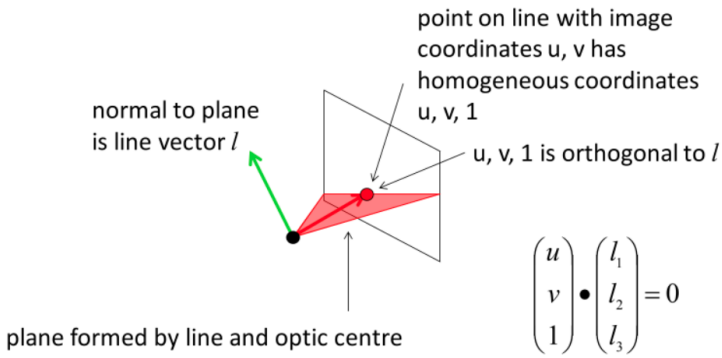

Figure 6: The line vector is normal to the plan formed by the line in an image and the optic centre of the image.

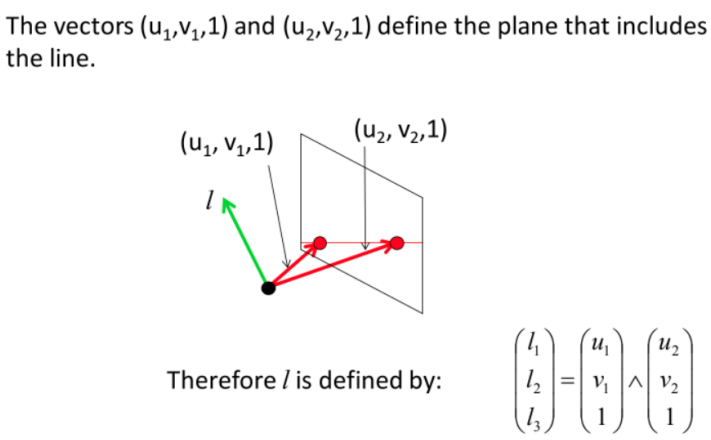

Figure 7: The line vector is normal to the plan formed by the line in an image and the optic centre of the image.